

Ideation Phase – Brainstorm & Idea Prioritization Template

Date: 25 June 2026

Team ID: XXXXXX

Project Name: Smart Lender

Maximum Marks: 4 Marks

Project Overview

Smart Lender is an intelligent loan approval prediction system that uses Machine Learning to assist banks and financial institutions in making faster, accurate, and data-driven lending decisions. The system analyses applicant information such as income, employment status, education, credit history, loan amount, and property area to predict whether a loan should be approved.

The objective of this project is to minimize manual verification, reduce loan default risks, improve decision-making efficiency, and provide quick and reliable loan approval predictions. The application combines a user-friendly web interface with a trained Random Forest Machine Learning model to automate the loan evaluation process.

Step 1: Team Gathering, Collaboration and Problem Statement Selection

Problem Statement

Traditional loan approval processes are often time-consuming, involve extensive manual verification, and may lead to inconsistent decisions. Financial institutions require an automated system that can accurately evaluate loan applications based on historical data.

Team Discussion

The team discussed several real-world banking challenges and identified loan approval prediction as a practical problem where Machine Learning can improve efficiency and accuracy.

Final Problem Selected

Smart Lender – Loan Approval Prediction using Machine Learning

Step 2: Brainstorming, Idea Listing and Grouping

Ideas Generated

- Predict loan approval using Machine Learning.
- Build a responsive web application using Flask.
- Train a Random Forest classifier for higher accuracy.
- Collect applicant details through an online form.
- Display loan approval or rejection instantly.
- Reduce paperwork and manual verification.
- Improve decision consistency using historical data.

- Make the application simple and user-friendly.

Grouping of Ideas

Machine Learning

- Data preprocessing
- Model training
- Prediction

Web Development

- HTML
- CSS
- Flask Backend

User Features

- Easy application form
- Instant prediction
- Responsive interface

Step 3: Idea Prioritization

Idea	Priority	Reason
Loan Approval Prediction	High	Core functionality of the project
Random Forest Model	High	Provides accurate predictions
Flask Web Application	High	Connects frontend with backend
Responsive User Interface	Medium	Improves user experience
Prediction Result Page	High	Displays final decision clearly
Future Database Integration	Low	Can be added in future versions

Final Selected Solution

Develop a **Smart Lender** web application using **Flask, Python**, and a **Random Forest Machine Learning model** to predict loan approval based on applicant information. The application provides instant results through an easy-to-use interface and helps financial institutions make faster and more reliable lending decisions.